

IEEE AESH Hackathon — Sustainability In Space

Concept Document — Core Sections

Problem

CubeSats are crippled by their own size, limited solar panels and small batteries leave them dangerously exposed during orbital eclipses, where idle sensors drain reserves the satellite can't recover. This means failed missions and, more critically, unresponsive satellites that become uncontrolled debris in increasingly congested orbits.

The downstream cost is real: a CubeSat monitoring Earth's albedo that loses power loses the very data tracking how declining surface reflectivity accelerates global warming a feedback loop we can't afford to go blind on.

Solution

Our solution is an Autonomous Power Management System that acts as a survival instinct for the CubeSat dynamically syncing its workload to the solar cycle so batteries never drain to failure. As the satellite moves between sunlight and eclipse, the system automatically powers hardware up or down based on what's actually needed at that orbital position.

What sets it apart is the distinction between active and passive power management, combined with dual-purpose hardware design squeezing more function out of fewer components to keep the power budget lean without sacrificing mission capability.

Architecture

During sunlit passes, the ESP32 runs continuously with the MOSFET open and the VEML7700 actively collecting albedo data. When the solar panel output drops signaling orbital eclipse the ESP32 fires the MOSFET to cut the sensor rail and shifts into a 60-second deep-sleep polling cycle, waking briefly to check whether solar generation has resumed or temperature has exceeded 120°C. Once sunlight is restored, the system exits the sleep cycle and returns to full active sampling.

Limitations

The transition from active sampling to sleep mode depends on the solar panel signal dropping, introducing a short lag before the MOSFET responds. The 60-second polling interval also means sunlight restoration is detected with up to a 60-second delay, slightly shortening each usable sampling window. Hardware remains COTS and unshielded against radiation, and the MOSFET is still a single point of failure with no software fallback.

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Evidence

Power consumption was measured and modeled across a full 100-minute LEO orbit (70 min sunlit, 30 min eclipse). At baseline, the always-on system draws 40.0086 mA continuously. With eclipse-gated power management active, eclipse-phase draw drops to 0.0111 mA a reduction of over 99% during that window yielding a total energy saving of 0.066 Wh per orbit and 0.95 Wh per day across 14.4 daily orbits. Overall, the optimized system consumes 30% less power and extends battery life by 42% compared to the always-on baseline.

Impact

By recovering 0.95 Wh per day and extending battery life by 42%, the system directly reduces the rate of premature mission failures the leading cause of uncontrolled debris in LEO. For student teams and low-budget operators, who make up the majority of CubeSat launches, this margin is often the difference between a completed mission and a dead satellite. Keeping these satellites operational longer means more continuous albedo data feeding into climate models that track Earth's declining reflectivity a feedback loop with direct implications for global warming projections. The solution requires no additional hardware or launch cost, making it immediately deployable across the growing CubeSat ecosystem.

Validation & Metrics

- **Primary metric:** 30% overall power reduction and 42% battery life extension per satellite, validated across a simulated 100-minute LEO orbit model.
- **Secondary metrics:** No added hardware cost or mass. The MOSFET and sensors are COTS components already common in CubeSat builds. Eclipse-phase draw cut from 40.0086 mA to 0.0111 mA.

Tools & Resources

- **Inputs/Sensors:** VEML7700 (albedo/light sensor) + LM35 (temperature sensor)
 - **Processing:** ESP32 microcontroller running C++ firmware — 60-second deep-sleep polling cycle
 - **Gatekeeper:** P-Channel MOSFET — physically cuts the power rail to the sensor array
 - **Simulation:** Python (orbit power simulation), Proteus (circuit simulation & validation), Velxio (system modeling).
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